

Supplementary Materials for “Underwater Organism Color Enhancement via Color Code Decomposition, Adaptation and Interpolation”

I. MORE VISUALIZATIONS

A. More Color Enhancement Results

Fig. 1, Fig. 2 and Fig. 3 show the enhanced images on EUVP (image-net) [2], LSUI [3] and UFO-120 [1] datasets. The visual results demonstrate that the proposed ColorCode can achieve competitive performance. The details, textures and colors are effectively enhanced and corrected. The color enhancement capability of ColorCode is further verified.

B. More Color Fine-tuning Results

Fig. 4, Fig. 5, Fig. 6, Fig. 7, Fig. 8 and Fig. 9 show the color fine-tuning results on multiple scenes. The results show that the colors of underwater organisms can be fine-tuned when images with long-wavelength colors are used as guidance. The guidance can be underwater images or natural images. The color fine-tuning capability of ColorCode is further verified.

C. More Color Interpolation Results

Fig. 10, Fig. 11, Fig. 12 and Fig. 13 show the color interpolation results for multiple underwater organisms. The interpolated image shows that color diversity can be effectively enriched through color code sampling. The color interpolation capability of ColorCode is further verified.

REFERENCES

- [1] M. J. Islam, P. Luo, and J. Sattar. Simultaneous enhancement and super-resolution of underwater imagery for improved visual perception. In *16th Robotics: Science and Systems, RSS 2020*, 2020.
- [2] M. J. Islam, Y. Xia, and J. Sattar. Fast underwater image enhancement for improved visual perception. *IEEE Robotics and Automation Letters*, 5(2):3227–3234, 2020.
- [3] L. Peng, C. Zhu, and L. Bian. U-shape transformer for underwater image enhancement. *IEEE Transactions on Image Processing*, 2023.

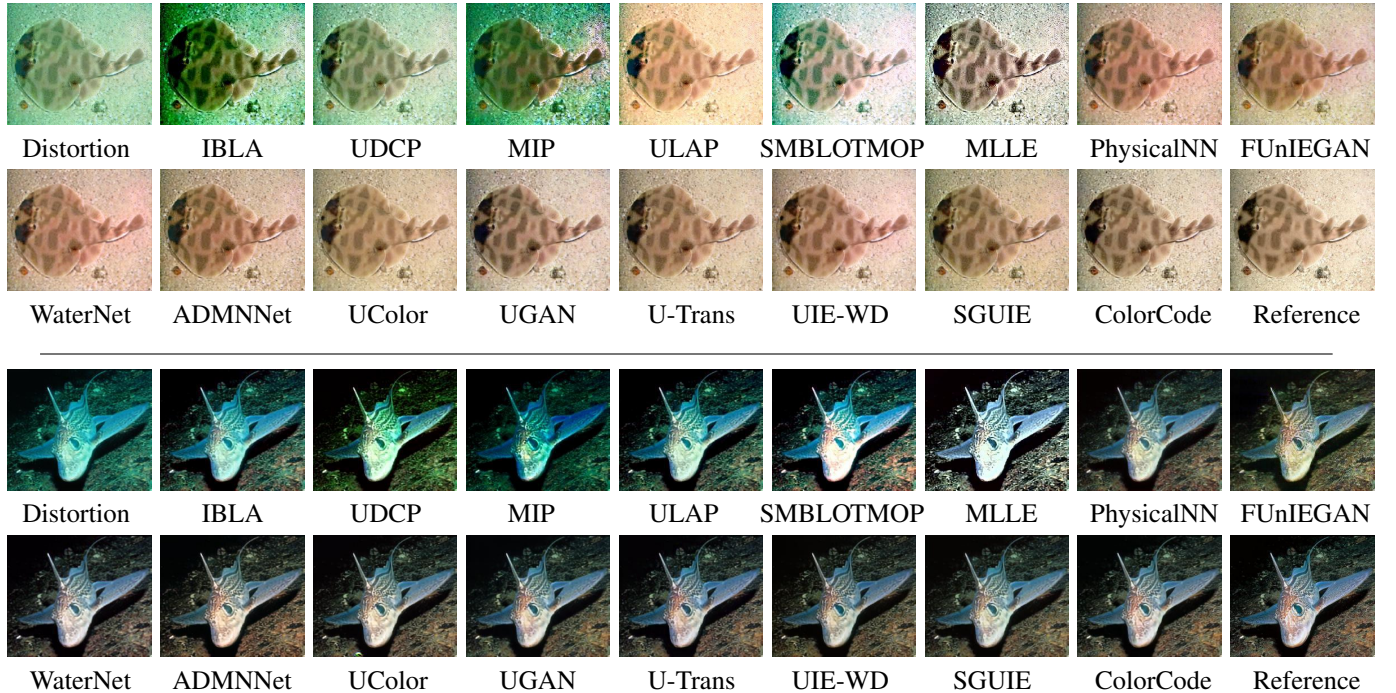


Fig. 1. Visual results obtained by various UIE algorithms on EUVP-I dataset.

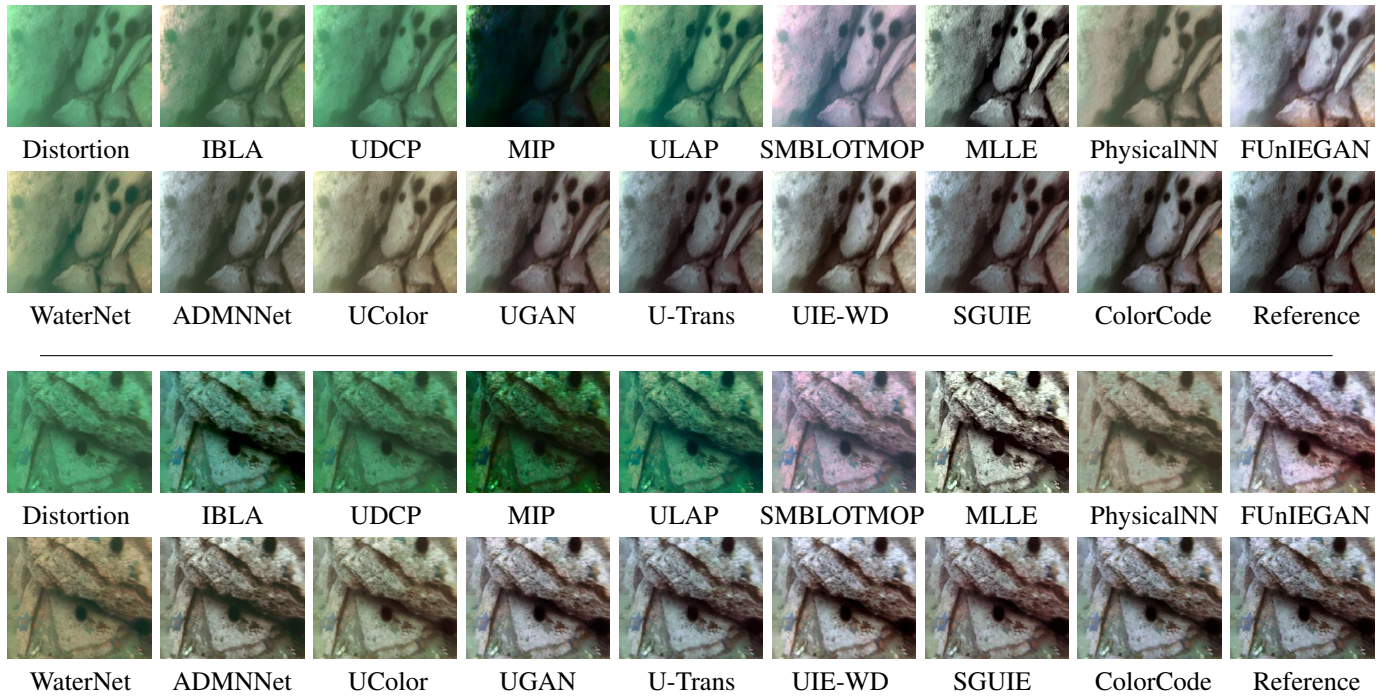


Fig. 2. Visual results obtained by various UIE algorithms on LSUI dataset.

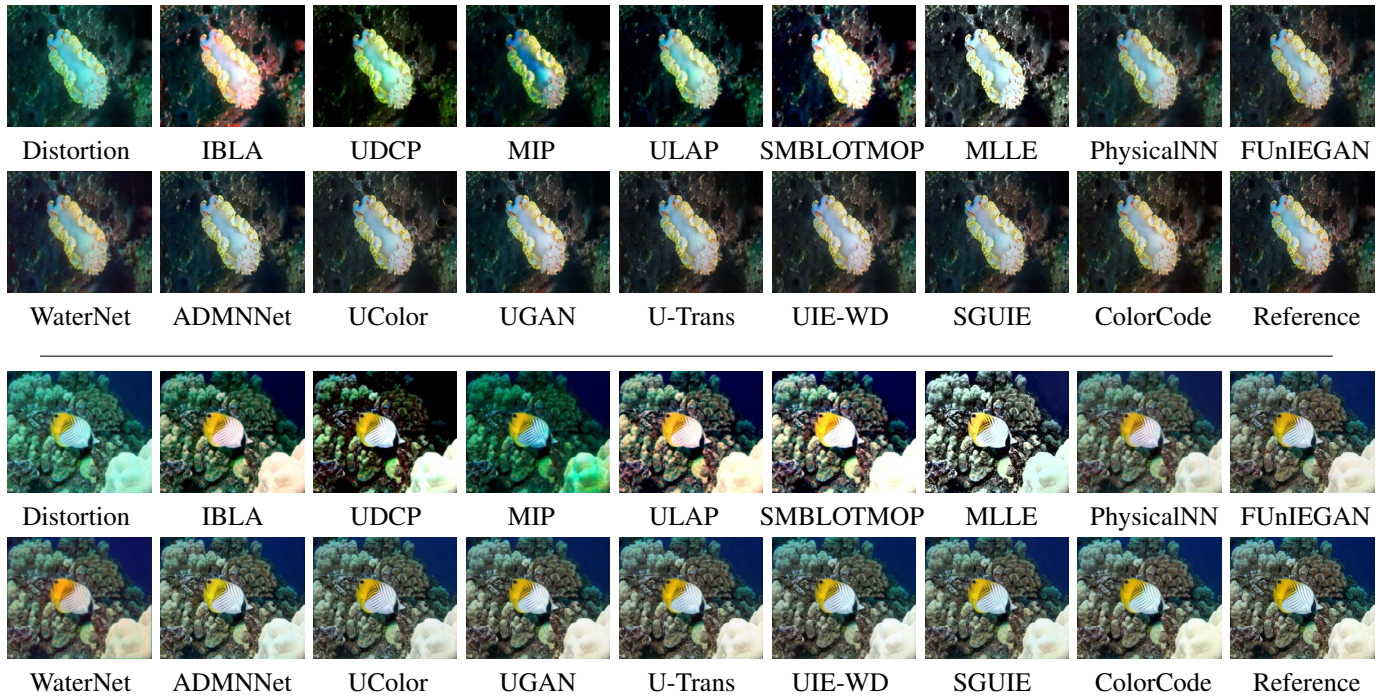


Fig. 3. Visual results obtained by various UIE algorithms on UFO120 dataset.

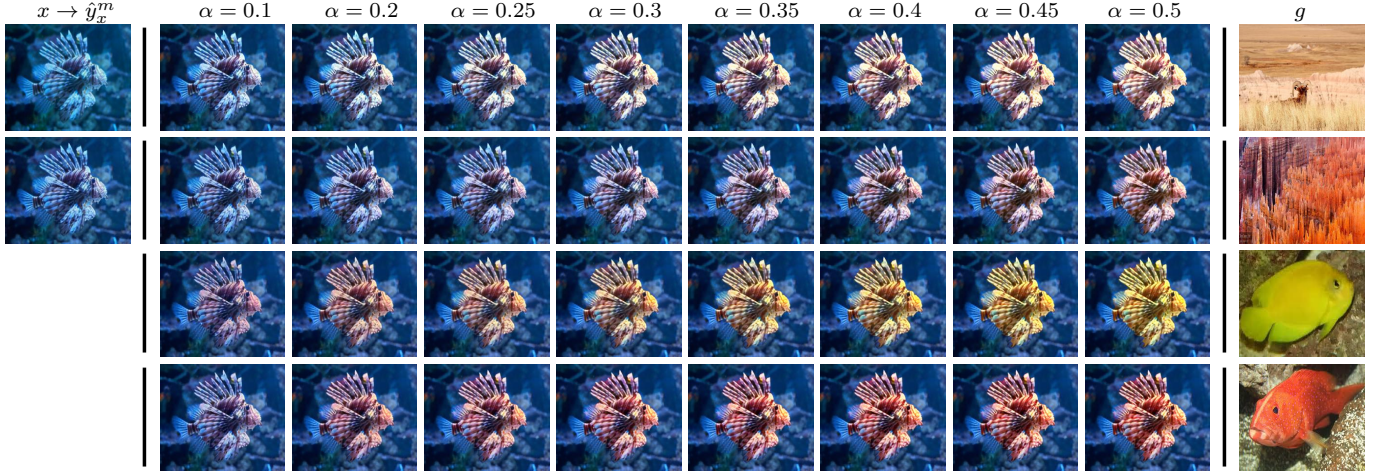


Fig. 4. Visualization of ColorCode's color fine-tuning capability. The α represents the weight parameter of the color guidance function $f(x, y, \alpha)$. For the first image of every two rows, the original distorted underwater image x and original enhanced result $\hat{y}_x^m$ are images at the top and bottom, respectively.

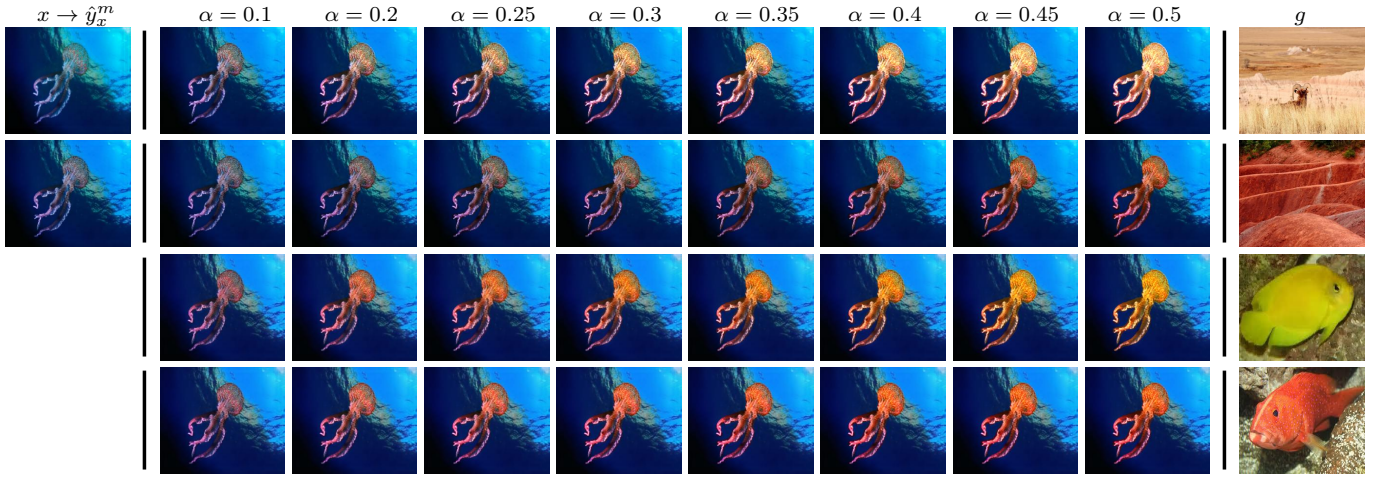


Fig. 5. Visualization of ColorCode's color fine-tuning capability. The α represents the weight parameter of the color guidance function $f(x, y, \alpha)$. For the first image of every two rows, the original distorted underwater image x and original enhanced result $\hat{y}_x^m$ are images at the top and bottom, respectively.

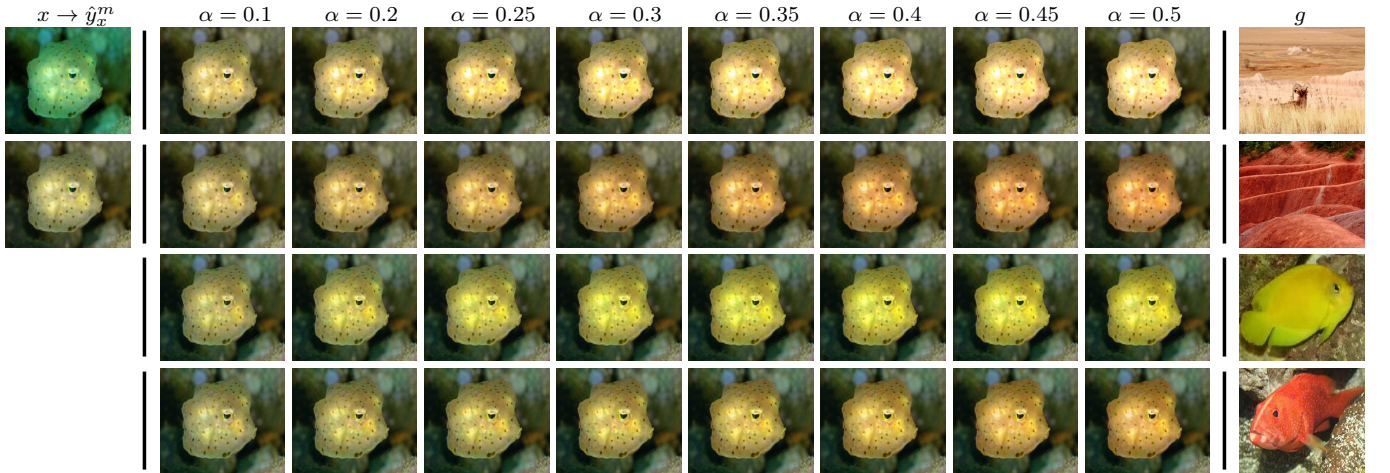


Fig. 6. Visualization of ColorCode's color fine-tuning capability. The α represents the weight parameter of the color guidance function $f(x, y, \alpha)$. For the first image of every two rows, the original distorted underwater image x and original enhanced result $\hat{y}_x^m$ are images at the top and bottom, respectively.

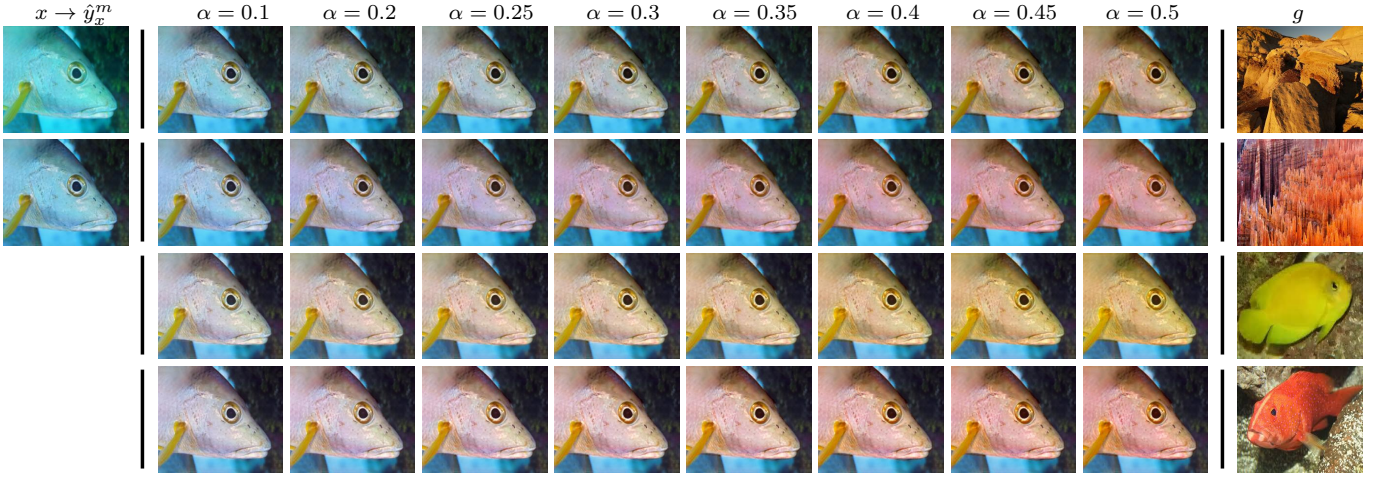


Fig. 7. Visualization of ColorCode's color fine-tuning capability. The α represents the weight parameter of the color guidance function $f(x, y, \alpha)$. For the first image of every two rows, the original distorted underwater image x and original enhanced result $\hat{y}_x^m$ are images at the top and bottom, respectively.

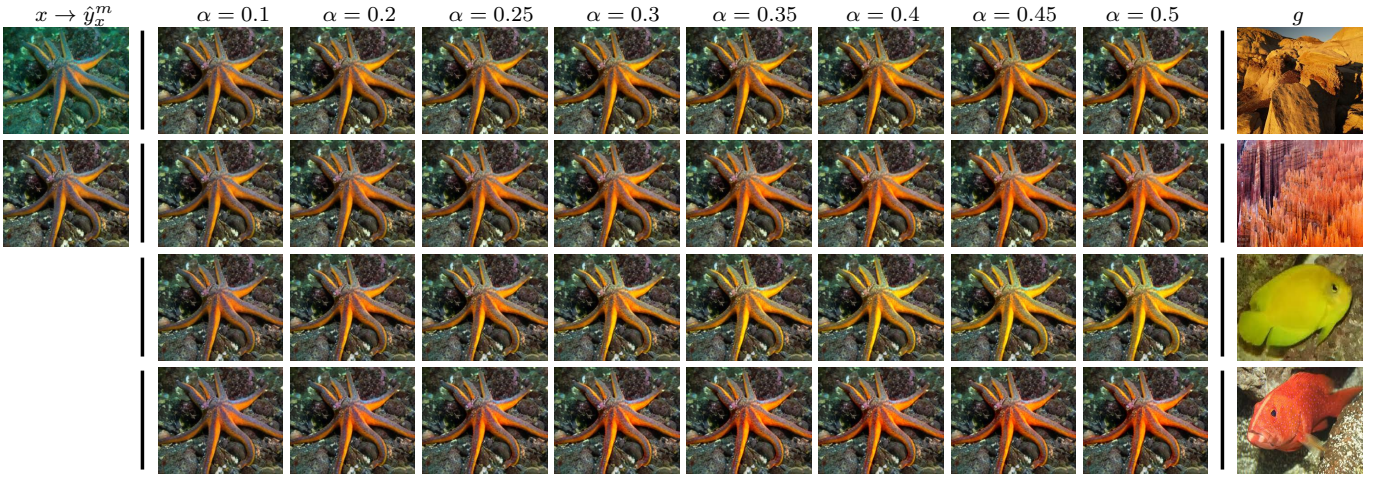


Fig. 8. Visualization of ColorCode's color fine-tuning capability. The α represents the weight parameter of the color guidance function $f(x, y, \alpha)$. For the first image of every two rows, the original distorted underwater image x and original enhanced result $\hat{y}_x^m$ are images at the top and bottom, respectively.

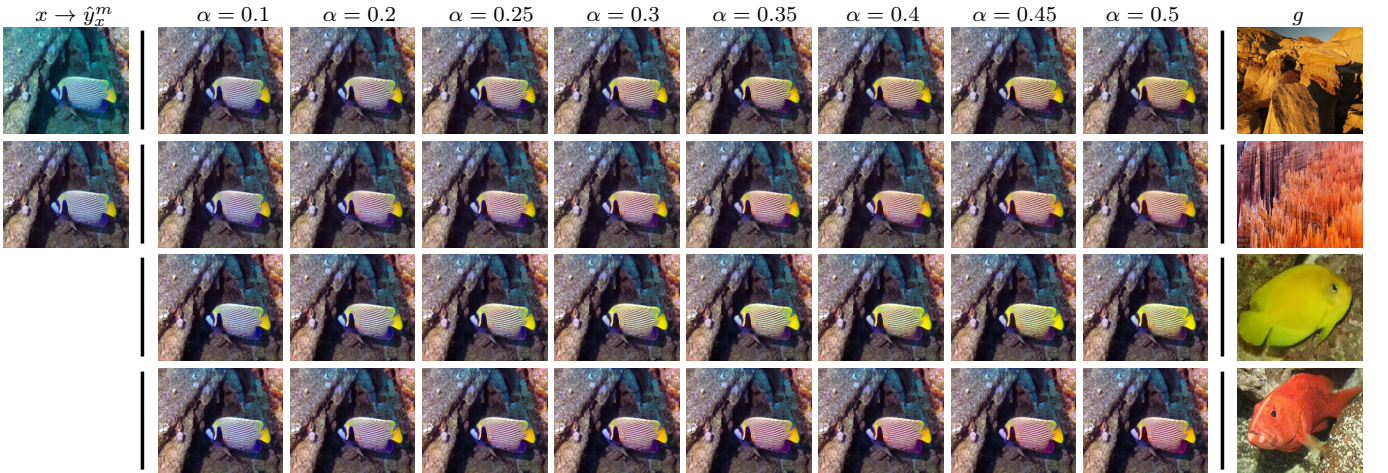


Fig. 9. Visualization of ColorCode's color fine-tuning capability. The α represents the weight parameter of the color guidance function $f(x, y, \alpha)$. For the first image of every two rows, the original distorted underwater image x and original enhanced result $\hat{y}_x^m$ are images at the top and bottom, respectively.

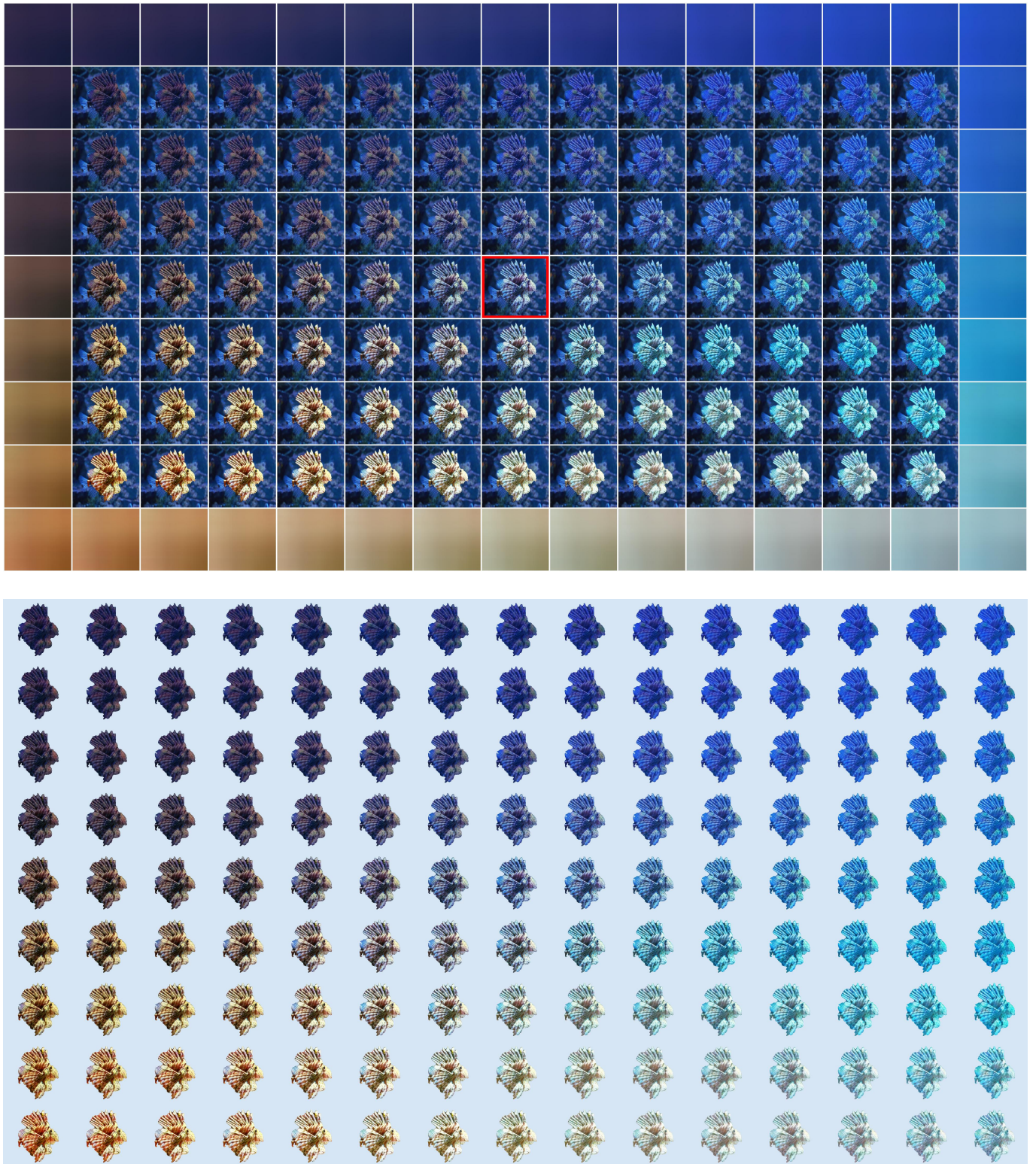


Fig. 10. Diversification results obtained by sampling color codes.

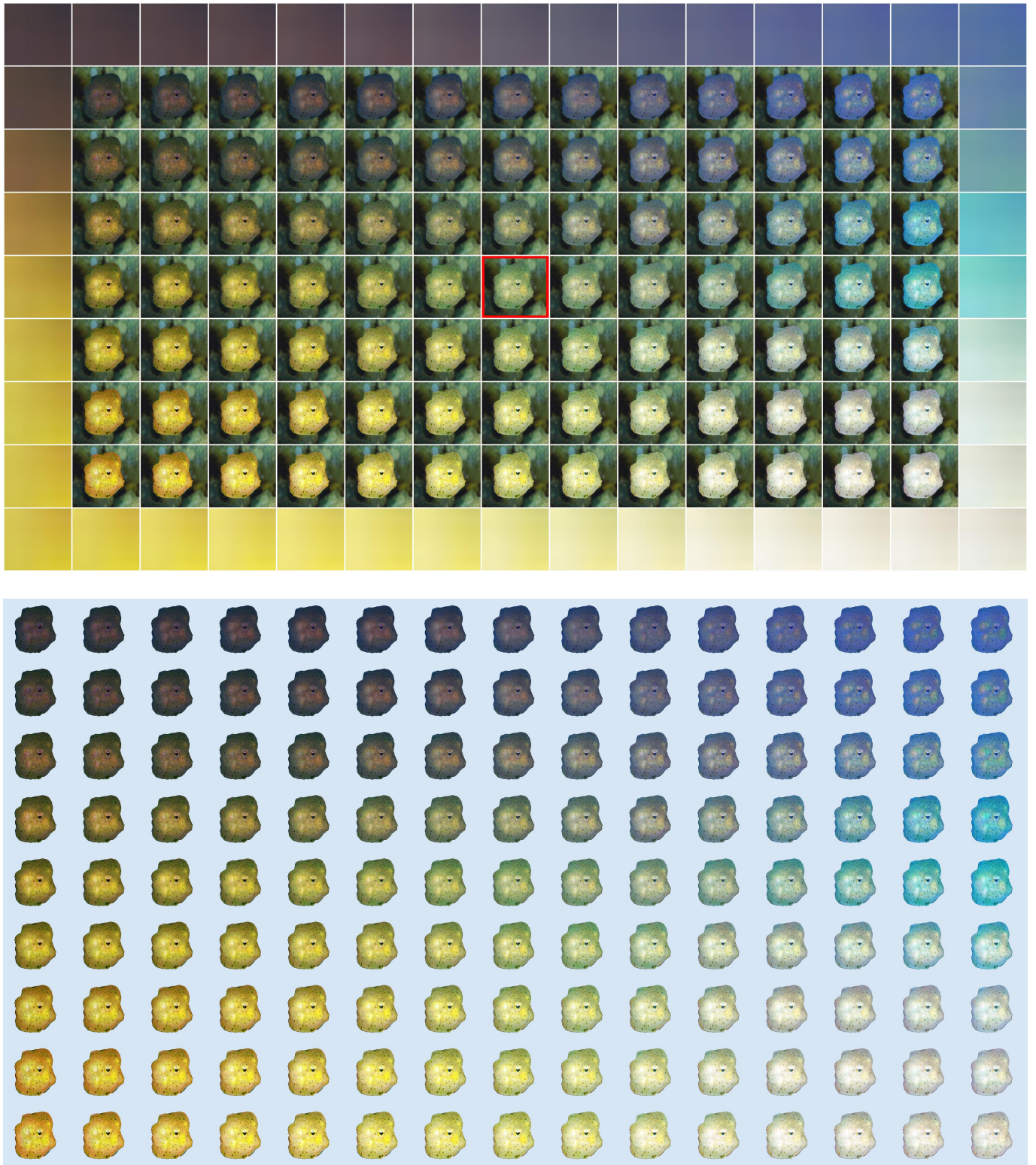


Fig. 11. Diversification results obtained by sampling color codes.

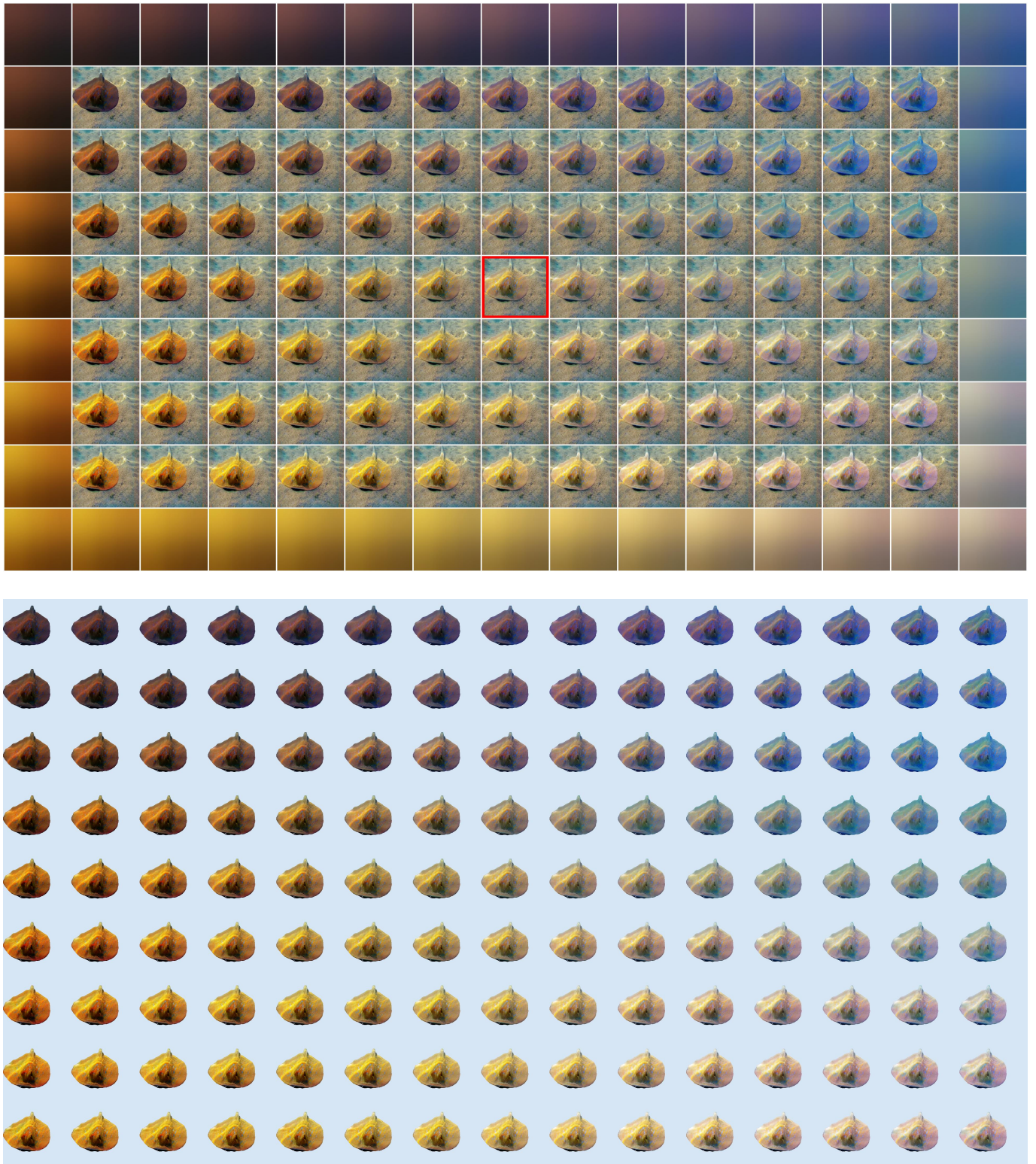


Fig. 12. Diversification results obtained by sampling color codes.

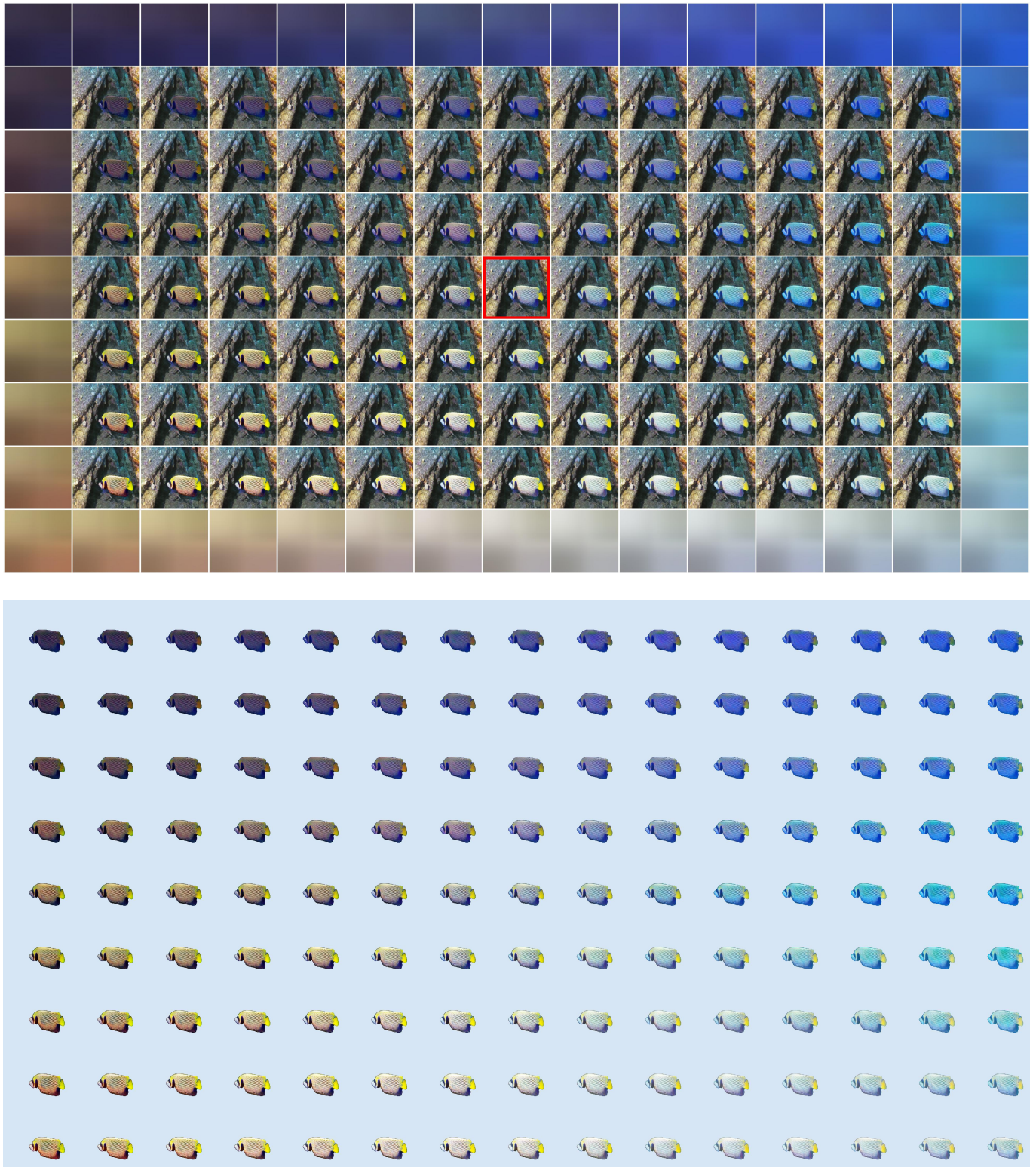


Fig. 13. Diversification results obtained by sampling color codes.